

Don't Solve Math in Your Head

Method over talent: school math for grades 10-11 and the 2028 CSAT

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Why I Came Back from Cambridge for a Grade-10 Midterm

University of Cambridge, Mathematics (BA, MA) · College Scholar (awarded for academic distinction).

Yonsei University: first-round admission to Dentistry (2022) and Pharmacy (2024) through the essay-based track.

People hearing my background for the first time usually ask the same question. “With a record like that, shouldn’t you be teaching the top students in their final year, or a medical-school essay class? Why, of all things, grade 10 and 11 school exams?”

The answer is simple. I wanted to prove something.

Among people gathered from every corner of the world, people who really had been “born with a head for math,” what I discovered was, unexpectedly, the exact opposite of what I had expected. The way a person approached a problem — their method and attitude — mattered as much as the ability they started with. I wanted to find out with my own hands whether that single principle would hold just as well in the unfamiliar, unforgiving world of Korean entrance-exam mathematics.

So I, by then a teacher, became a candidate again and sat down in front of the very same paper as that year’s high-school seniors. And against odds of more than a hundred to one, I was admitted in the first round to Yonsei University’s dentistry and pharmacy programmes, one after the other, through the essay-based track.

Not because I am some born genius. I had simply worked out, precisely, the pattern of decoding that Korean math exams demand. Whether it is pure mathematics at Cambridge or the school exam you will sit tomorrow, mathematics in the end is not a talent. It is a craft of decoding, and a craft is won by training.

I know this: at this very moment, countless tenth- and eleventh-graders are sitting at their desks until eleven at night, swallowing the unfairness of it all. They have worked through three problem books, kept an error notebook, sat through every online lecture that is supposed to be the best, and this midterm score is exactly the same as the last one. And the parents who watch them from behind are eaten up with worry.

Are you, right now, blaming yourself for not having been born with a head for math?

A stalled score is not a reason to question your ability or effort first. Before adding more hours, look at how you spend them. Can you solve a question unaided after thinking you understood it? Do you keep getting stuck at the same point?

“In the end, math is all about the head.” “A head for math is something you are born with.” I am convinced these sayings are half a lie. What looked like talent turns out, on inspection, to be a very precisely engineered set of habits.

That is why I focus on school mathematics in grades 10 and 11. Difficult final-year questions depend on concepts and habits learned earlier. Repair a gap revealed by the first grade-10 midterm, and later material becomes easier to take in. This does not mean the contest is over early. It means there are things you can change early.

To the parents who have opened this book, a word as well. I know what it feels like when the hours your child spends at the desk keep growing and the grades do not move. Should we add another private academy? Is our child just not built for math? Please do not rush to that conclusion. Before adding another academy, look with your child at where the current work is getting stuck. At the end of every chapter, and in the final chapter, Chapter 8, this book speaks frankly about the parent’s role too: how to stand firmly beside your child while that method takes hold.

Take the method written in this book and apply it for one semester, not in the way you are used to but exactly as this book lays it out. Replace the habit of following with your eyes with the habit of writing it down by hand. Start tracing the examiner’s intent backwards, not only in the problems you got wrong, but also in the ones you got right.

Drop the wrong method, get onto the right track, and you will be able to see, step by step, what has changed. Check the process first, before you check the score.

— Sungmin Kim

Why Middle-School Math Stars Collapse in High School

Opening Story: Hyunwoo's Year

"Hyunwoo" below is a fictional name. To protect privacy, his story is a composite drawn from several of my students.

All through middle school, Hyunwoo's math report cards always showed scores in the nineties. His parents, and Hyunwoo himself, never doubted that he was "at least decent at math." Then came the first midterm of high school, and the result he was handed shook that belief to its foundations. Grade 4. On the five-band scale Korean schools use, that is the second-lowest band: a number he had never once seen in his life.

Do not misunderstand. Hyunwoo was not a lazy kid. If anything, he was more diligent than anyone. He worked through problem books by the stack and never once fell behind on his academy homework. So why did he collapse?

Three days of watching him study was enough to see the cause. Hyunwoo was not solving problems; he was watching them. He would follow the worked solution with his eyes, nod, "ah, so that's how it goes," and never write a single line of it again with his own hand. When he got a problem right, he moved on without asking why. Concepts he had memorised, not understood. That approach held up through middle school, but high-school math drove straight into exactly that gap.

One scene in particular stays with me. One day I asked Hyunwoo to solve, on a blank page, a problem he had just told me he "understood." His pencil stopped at the second line. This was a problem he had nodded through, solution in hand, five minutes earlier. "Um... what did I do here again?" He laughed, embarrassed, but in that moment I was certain. Hyunwoo's problem was not his head. It was a habit: he had never learned to tell studying by watching from studying by solving. There is never just one Hyunwoo in a classroom. If anything, the more diligent the student, the deeper this trap tends to run.

We changed exactly one thing: the decoding habit this book is about to describe. What he used to solve by eye, he now wrote out by hand. Instead of the problems he got wrong, he traced backwards the examiner's intent in the problems he got right. For the first month Hyunwoo was frustrated; the pace

looked slow. But after one semester he was at Grade 3, and one semester after that, Grade 2. Looking back at that first Grade 4, it was a change Hyunwoo himself found hard to believe.

Hyunwoo was no special genius. His talent had not suddenly grown. He had simply dropped a wrong method and switched to a right one.

What changed Hyunwoo? From here on, let us dissect, one by one, the traps that quietly bring down so many ordinary students, and the craft of decoding that gets them out.

1. “I Know All This, So Why Do I Keep Getting It Wrong?” — Why Middle-School Studying Collapses in High School

Hyunwoo’s story is not an unusual event. Every March, dozens of students like him sit down in front of me. Kids who were told all through middle school that they were “pretty good at math” collapse in front of their first high-school exam, all wearing the same expression. There is something to check first. **Can they use what they learned in middle school when a high-school problem combines longer conditions and several concepts?**

Miss this difference and all the diligence in the world just spins in place. Let me break it into three parts.

From Studying by Volume to Studying by Logic

Middle-school math can be conquered by volume. The exam scope is narrow and the problem types that appear are limited. So if you cycle through three or four problem books, what students call “grinding,” and drill the types you keep missing, the score usually follows. This is exactly the approach that worked for Hyunwoo in middle school.

Take a simple example. In middle school, “solve the quadratic equation” meant you factorised, or plugged into the quadratic formula, and you were done. One question, one road. Arrive in high school and the same quadratic turns into this:

Find the range of real values of k for which the quadratic equation $x^2 - 2kx + k + 2 = 0$ has two distinct real roots.

The ability to “solve” an equation is barely used here. The real test is whether you can translate the condition “two distinct real roots” into the discriminant condition $D > 0$. The calculation after that is a minor finishing touch.

This is how high-school math works: it pretends to ask about one concept while actually asking about the connections between concepts. Memorise as many problem types as you like; unless you have trained the logic that ties those types together, you will freeze in front of the first problem you have

never seen before.

The Expiry Date on Memorising Problem Types

Up through middle school, memorising types, “when this kind of problem appears, do this,” is a powerful weapon, because the problems wear more or less fixed faces. But the examiners who set high-school papers ask about the same concept with a different face every time. The solution you memorised yesterday and the problem on today’s paper no longer match one to one.

I still remember what Hyunwoo said the first time he came to me.

“Sir, I definitely know all of this. But the moment I get the exam paper, I’m stuck at ‘how do I even start this?’”

The moment those words come out is the signal that middle-school studying has passed its expiry date. “Knowing it but not being able to start” is not a shortage of ability. It means the knowledge is stacked in your head as information you have heard of, not as tools for solving problems. What you need is not more types. It is **the power to rebuild, by yourself, the principle hiding behind the type.**

The Cliff Between “I Understand It” and “I Can Solve It”

When Hyunwoo looked at a worked solution and nodded, “ah, right, this one,” that was not understanding. It was assent. He had followed a road someone else had cut and acknowledged that it made sense. Assent is easy, it feels good, and that is precisely what makes it dangerous: it gives you the illusion that the studying is done.

Try an experiment. Take a problem whose solution you have just read and felt you “understood.” Cover the solution completely and solve it alone, from the beginning, on a blank page. Notice where your hand stops. That is where to revisit the explanation or check the concept.

In the exam room nobody cuts the road for you. You have to write the first line yourself, on a blank page. Between “I understand it” and “I can solve it with my own hand” lies a cliff deeper than most people expect. Most students, unaware that the cliff even exists, believe “I understand it” right up until the exam, and then they fall.

2. Examiners Dig Traps, and Every Course Sets a Different One

A large share of the problems students miss in high-school math are missed not because they did not know, but because they were caught. The examiner knows exactly where a student is likely to misstep and sets the trap right there. So our job is not to solve problems faster. It is **to see where the trap is**

first, through the examiner's eyes. And the traps look different in every course.

Common Mathematics 1 and 2: The One-Line Condition Trap

What students most often overlook in Common Mathematics is the one-line condition inside the problem. Short phrases like “has two distinct real roots,” “for all real numbers x ,” or “ x is a natural number” in fact decide the direction of the whole problem.

Look again at the problem above. “Two distinct real roots” means the discriminant must be greater than zero ($D > 0$). But had the problem said “has a repeated root,” the condition would be $D = 0$, and had it said “has real roots,” $D \geq 0$, and the answer changes completely. A small change in wording changes the answer. The student who misses that line is not getting it wrong despite knowing it; he is getting it wrong **because he did not read the condition**. There is one thing I make every student do: as you read the problem, circle every word that is a condition. This one small habit visibly cuts the mistakes in Common Mathematics.

Algebra and Calculus I: The Definition Trap

The traps in these courses test how precisely you hold on to a concept's definition. A student who skimmed past the domain of a function collapses the moment a problem asks whether the function is even defined at that value. A student who knows only by feel what it means for a limit to “exist” gets caught on a problem where the left-hand and right-hand limits disagree.

For example, the single sentence “the function $f(x)$ is continuous at $x = 1$ ” actually demands three things at once: ① the function has a value at $x = 1$, ② the limit exists at $x = 1$, and ③ the two are equal. A student who can state these three conditions as sentences avoids the trap. A student who knows by feel that “if it's roughly joined up, it's continuous” walks straight into the trap the examiner left at the edge of the definition. Differentiability, continuity, increasing and decreasing: hold the definitions of these words as exact sentences, not as feelings. Algebra is no different. Missing the condition on a logarithm, that the base is a positive number other than 1 and the argument is positive, or forgetting the case where the common ratio is 1 in the sum of a geometric sequence: these mistakes are precisely the definition trap.

Probability and Statistics: The Translation Trap

In Probability and Statistics the trap is not in the calculation but in the translation. At the very moment a situation written in words is turned into an expression, a condition leaks out without a sound. The classic example is the phrase “at least.”

“The probability that at least one is defective”

Try to compute this sentence head-on and the cases branch into a tangle. Know how to translate “at least one” into “everything, minus the case where none is defective” (the complementary event), and it is over in one line. Failing to distinguish “at the same time” from “one after another,” or missing whether what you drew is put back (with replacement) or not, are mistakes of the same kind. Open the error notebook of any student in Probability and Statistics and the entries for “translated the sentence wrongly” outnumber the calculation slips by a wide margin.

The traps in these three courses look different, but they share one root: **the examiner always aims at the spot where a student will read carelessly and move on.** Equip yourself with that one point of view, and with the same ability you will get more problems right.

3. The Illusion of Knowing a Concept: How Top Students Actually Study Concepts

If you have read this far and concluded “so the point is that concepts matter,” you are half right. The trouble is that almost every student believes “I know the concepts.” The real difference is not whether you know a concept but how deeply you know it. Top students take hold of a concept in three steps.

Step 1: Write the Definition on a Blank Page, in Your Own Words

Reading a textbook definition with your eyes and closing the book to rewrite it in your own words on a blank page are two completely different things. Do the second and you can distinguish a familiar sentence from a concept you can explain yourself. If you cannot rebuild the definition on your own, that concept is not yet yours.

Step 2: Derive the “Why” by Hand

Memorise a formula accurately, but do not stop there: learn how to derive it too. Take the quadratic formula. A student who simply memorised it and a student who once divided both sides of $ax^2 + bx + c = 0$ by a , completed the square, and built the formula with his own hands become entirely different people in the exam room. For the first, forgetting the formula is the end. The second, having forgotten it, rebuilds it on the spot and carries on. Better still, because deriving it taught him in his bones what the tool called “completing the square” is for, other problems that need that tool start to open up as well. Dig into the question “why?” just five times and you can see a concept’s roots.

Step 3: Explain It to Someone, Then Ask the Question Backwards

The most powerful check is explanation. Teach the concept to the friend in the next seat, or to an empty room if you have to. Wherever you get stuck is exactly what you do not know. Then add one more thing. After solving a problem, ask the question in reverse: “Why did the examiner put this

condition in?" Once this backwards question becomes a habit, you will find that the trap is the first thing you see when you look at a problem.

These three steps are slow. At first the pace is slow enough to be maddening; that is how Hyunwoo spent his first month. But a concept packed down this way does not give way under exam pressure. A tower built fast falls fast. Ground packed slowly holds to the end.

ACTION CHECK

Three Things to Change Today from This Chapter

1. **Circle the conditions in the problem.** Do not just read words like "distinct," "all," and "natural number" with your eyes; mark them by hand.
2. **Cover the solution and solve on a blank page.** The more a problem feels "understood," the more it needs to be solved again, alone, from the start, with the solution covered. Where your hand stops is your real weakness.
3. **Ask "why?" once about a problem you got right.** Do not look only at the ones you missed. Ask once, of a problem you got right, "why did the examiner set it this way?"

WORKSHEET

Self-Check: The "I Understood It" Trap

Knowing and being able to solve are different things. Check yourself right now with one problem you have just studied. Honest ticks are the whole point of this diagnosis.

Pick one problem you recently felt you "understood," cover the solution completely, and tick the boxes below.

- ☐ I could start the first line by myself
- ☐ I reached the end without getting stuck
- ☐ I can explain in words why that formula or concept is used
- ☐ My answer was right, and I can explain why it is right

If even one box is unticked, that is the exact spot where you mistook assent for understanding. Write it down below and study it again.

Where I got stuck _____

Concept to restudy _____

One Page for Parents

- Q.** My child says “I understand it,” then gets it wrong on every test. Why?
- A.** “I understand it” usually means the child has looked at a worked solution and nodded. It does not mean the child can solve the problem alone on a blank page. Ask, “Can you explain this problem to me?” Wherever the explanation stalls is what your child actually does not know. Rather than checking whether the answer was right, check whether the child can explain it unaided.
- Q.** My child works through several problem books and the grades still don't move. Should I make them do more?
- A.** The problem is most likely the method, not the amount. Three problems written out by hand to the end, with the child explaining why each step is taken, are worth far more than ten problems solved by eye. “Can you explain just one of the problems you solved today?” is a better question than “How many problems did you do?”

In Closing: It Was the Method That Failed, Not the Talent

Let us go back to Hyunwoo. What changed him was not a head that suddenly got better. It was not more hours of study either. Only one thing changed: **the way he approached a problem.**

Once he set down the middle-school habits of grinding and memorising types, began rebuilding concepts in his own words, and started reading traps through the examiner's eyes, the grades moved, with the same head and the same hours. So if the number on your report card has you in despair right now, please do not mistake it for a verdict on your talent. It is nothing more than a signal to change your method.

In the next chapter we take out the first tool in the craft of decoding: the most concrete, fastest-acting way to turn math solved by eye into math solved by hand.

Solve by Hand, Not by Eye: How Top Students Design a Solution

Opening Story: Until Hyunwoo Picked Up His Pencil

You will remember Hyunwoo from Chapter 1. The first thing I overhauled about the way he studied was not, in fact, the concepts he knew or the problem books he used. It was **the way he held his pencil**.

I was more than a little surprised when I first sat beside him and watched him solve a problem. Even after reading the whole question, Hyunwoo would sit perfectly still for a long while. His eyes moved between the question and the blank space, but his hand did not. In his head he would sketch out, “this goes here, that goes there,” until the moment it became complicated; then he would say, “I don’t know,” and turn to the worked solution. His pencil moved only briefly, when he copied down the answer.

Hyunwoo was a student who solved by eye. And this is what students who work hard, yet whose grades refuse to rise, most often have in common. This chapter gives the most concrete account of how to replace that habit with the habit of solving by hand.

1. Why Math Solved by Eye Falls Apart

Studying by eye is comfortable. Your hand does not hurt, you seem to move quickly, and, best of all, it feels as though you have solved something. That feeling is the trap.

When the Working Gets Longer, Put It on Paper

The amount of information the mind can hold at once is far smaller than most people imagine. You can manage two or three simple steps of calculation in your head. But the moment three or four conditions become entangled and the expressions grow longer, it becomes easier to lose track when everything stays in your head. Hold on to the expression you wrote earlier and you lose the next step; think about the next step and the earlier expression fades.

I once had a student who came to me after an exam, seething. “Sir, I knew this problem. I multiplied both sides of the inequality by -2 and forgot to reverse the inequality sign. That one mistake cost me five marks!” His frustration was understandable. It was not that he did not know the rule. But when I

looked at his paper, the blank space beside that question was unusually clean. He had tried to multiply by -2 , move the terms across, and simplify everything in his head, all at once. For one instant, the rule “reverse the inequality sign” was pushed aside. This is how students with exceptionally clean exam papers usually lose marks in baffling ways.

“Three lines” is a practice rule for putting work on paper, not a measured limit of the mind. When conditions and calculations begin to intertwine, **write the intermediate steps where you can check them.**

Your Hand Is the Mind’s “External Memory”

Writing with a pencil is not merely recording something. It is the act of setting down on paper information that would otherwise be swept out of your mind. Write the conditions down and the mind can redirect the effort it was spending on remembering them towards thinking. The difference between a student who writes and one who solves by eye is not whether one has a better head. It is **how sparingly each uses the mind’s limited capacity**. Top students do not handle harder problems because their heads are larger. They handle them because they use the external memory in their hands well.

The Moment Solving by Eye Collapses in the Exam Room

The whole cost of the habit comes due in the exam room. Under pressure, the mind has less working space than usual. Even a problem you could normally solve by eye leaves you unable to write the first line, because your hand has never been trained for it. This is what lies behind the familiar claim, “I could do it at home, but I froze in the exam.” Train yourself to solve by hand and, even under pressure, your hand moves first and pulls your mind after it. That is the moment habit fills the place talent leaves empty.

2. Do Not Start Solving as Soon as You Read: The Thirty Seconds Called “Solution Design”

Do you start calculating before you have read the whole question? This time, **pause and plan the solution**. In practice, allow roughly thirty seconds and write three things in the margin. As the habit settles, adjust the time to the difficulty of the question and the time remaining.

① What Is Given — ② What Is Required — ③ The Tool That Connects Them

As you read, write “given conditions” at the top of the blank space and “what I need to find” below. Then draw an arrow between them and ask, “which concept or formula will connect these two?” Those thirty seconds turn the work from “charging in blindly” into “walking a road whose direction I know.”

Take this question from Common Mathematics 1.

Find the distance between the two points where the graph of the quadratic function $y = x^2 - 4x + 3$ meets the x -axis.

A student who solves by eye imagines the graph and then stops. A student who designs by hand writes this:

- **Given:** $y = x^2 - 4x + 3$
- **Required:** the distance between the two points where it meets the x -axis
- **Connecting tool:** meets the x -axis $\rightarrow y = 0 \rightarrow$ the two roots of a quadratic equation

Once you have written that much, the solution is already half done. Set $y = 0$, and $x^2 - 4x + 3 = (x - 1)(x - 3) = 0$. The two points lie at $x = 1$ and $x = 3$, so the distance is 2. There was nothing difficult about the calculation. The difficulty was only **the helplessness of not knowing where to begin**. A blueprint removes that helplessness.

The Blueprint Shows Its Real Power in a Variant

The force of this habit becomes clear the moment a question is twisted even slightly. Suppose the same question changes to this:

The graph of the quadratic function $y = x^2 - 4x + k$ meets the x -axis at two distinct points. Find the range of real values of k .

A student who has memorised only the type panics: “Wait, this is different from the last one.” A student who draws a blueprint starts in exactly the same way. Write down what is given and what is required, then translate “meets at two distinct points” into a tool — **two distinct real roots $\Rightarrow$ discriminant $D > 0$** . Then $D = (-4)^2 - 4k = 16 - 4k > 0$, so $k < 4$. The face of the problem changed, but **the way we found the road was exactly the same**. That is the decisive difference between memorising types and designing a solution.

Nor does this blueprint work only in grade-10 Common Mathematics. It works identically in grade-11 Algebra. Faced with “find the range of x for which $\log_2(x - 1)$ is defined,” the question may look forbidding, but the blueprint immediately calls up the tool — **a logarithm is defined $\rightarrow$ the base is a positive number other than 1, and the argument is positive $\rightarrow x - 1 > 0 \rightarrow x > 1$** .

A limit problem in Calculus I is no different. Write down what is given and what is required, and ask “which definition do I need?” The course may change, but the way you find the road never does. One method running through several courses: that is the real power of the design habit.

The More Stuck You Are, the More You Return to Your Hand

When a student who solves by eye gets stuck, they wander around inside their head. A student who solves by hand returns to the blank space and writes down what has emerged so far and what remains. Gather the information in one place. It becomes easier to see whether you need to connect what you have or revisit a concept.

3. The Five Stages of Decoding: Taking One Problem All the Way Through

If “solve by hand” still sounds vague, follow the five stages below exactly. I call them **the Five Stages of Decoding**. They are the one fixed routine for approaching a problem that runs through this entire book. The goal is to make your hand move through this sequence whenever you meet any question in any course. This time, rather than merely describing the stages, we will work through one problem together from beginning to end.

The graph of the quadratic function $y = x^2 + 2x + k$ meets the x -axis at two distinct points A, B . Given that the length of the line segment AB is 4, find the real value of k .

Stage 1: Mark the conditions.

Do not just read them with your eyes and move on. Copy the conditions down by hand.

“ $y = x^2 + 2x + k$ / two distinct points A, B on the x -axis / $|AB| = 4$ / $k = ?$ ”

Stage 2: Call up the tool.

“Two points where the graph meets the x -axis” are the two roots when $y = 0$, and “the length of AB ” is the distance between those roots. The tool to call up here is the relationship between roots and coefficients. (Draw for yourself a parabola opening upwards and crossing the x -axis at two points; then your hand, not your eye, will see that AB is the difference between the roots.)

Stage 3: Set up the equation.

The intersections with the x -axis are found by setting $y = 0$; in other words, they are the two roots of $x^2 + 2x + k = 0$. Call the roots α and β . By the relationship between roots and coefficients, $\alpha + \beta = -2$, $\alpha\beta = k$. And $AB = |\alpha - \beta| = 4$.

Stage 4: Calculate.

Only now do we begin the calculation. Since $|\alpha - \beta|^2 = (\alpha + \beta)^2 - 4\alpha\beta$,

$$4^2 = (-2)^2 - 4k \rightarrow 16 = 4 - 4k \rightarrow 4k = -12 \rightarrow k = -3.$$

The answer is $k = -3$.

Stage 5: Check and trace back.

Getting an answer is not the end. Use the discriminant to check that it really does satisfy the condition “meets at two distinct points.” $D = 4 - 4k = 4 - 4(-3) = 16 > 0$. The roots are distinct, so the condition is satisfied. Then ask once more: “Why did the examiner give the condition about distance?” To make us use the relationship between roots and coefficients and the tool $(\alpha + \beta)^2 - 4\alpha\beta$. This act of tracing back becomes your ability the next time you meet a similar problem.

① Mark conditions

Circle every condition in the question



② Call up the tool

Translate conditions into concepts and formulae



③ Set up equations

Turn the target and conditions into equations



④ Calculate

Once the earlier stages stand, calculation is easy



⑤ Check and trace back

Check even a correct answer; add one line on the examiner's intent

When I first made Hyunwoo follow these five stages, he said, “This makes me far too slow.” He was right: it is slow at first. But after a month his hand began moving on its own, and he actually solved problems faster. When you know the road, you do not waste time wandering.

ACTION CHECK

Three Things to Change Today from This Chapter

1. **When you read a problem, first write “given / required” in the blank space.** Calculation comes afterwards.
2. **Do not solve more than three lines in your head.** Beyond three steps, put the work down on paper.
3. **Even after finding the answer, return to the conditions.** In particular, check conditions such as “two distinct roots” and “positive” against the discriminant or definition.

Practice Sheet: The Five Stages of Decoding

Do not solve in your head. Fill in the five spaces below by hand. First learn the flow from the worked example, then choose one new problem and complete the blank spaces yourself.

SAMPLE ENTRIES

Problem: Find the range of real values of k for which the quadratic equation $x^2 - 6x + k = 0$ has two distinct real roots.

- ① **Mark conditions** — two distinct real roots; the target is the range of k .
- ② **Call up the tool** — two distinct real roots $\Rightarrow$ discriminant $D > 0$.
- ③ **Set up equations** — $D = (-6)^2 - 4k > 0$.
- ④ **Calculate** — $36 - 4k > 0$, so $k < 9$.
- ⑤ **Check and trace back** — for example, when $k = 0$, the roots of $x^2 - 6x = 0$ are 0, 6, which are distinct.
The condition is satisfied.

Now use your own problem: Choose and write down one quadratic-equation or quadratic-function problem from your textbook or problem book.

Problem _____

- ① **Mark conditions** _____
- ② **Call up the tool** _____
- ③ **Set up equations** _____
- ④ **Calculate** _____
- ⑤ **Check and trace back** _____

One Page for Parents

Q. The blank space is always clean when my child solves a problem. Does that mean they are good at mental arithmetic?

A. Unfortunately, the opposite is more likely. Clean blank space usually marks a problem “solved” by eye. When you look at a marked exam paper, look at the blank space before the mark. Whether the process has been written down by hand tells you more than the mark does about your child’s room to improve.

Q. When I say, “write as you solve,” my child says it wastes time and refuses.

A. Of course it looks slow at first. But “solving quickly” and “solving correctly” are not the same, as the next exam marks will soon show. Ask your child to write everything out for just one month, and watch together how the blank space changes. That change will persuade them better than nagging will.

Q. My child’s handwriting is so poor that they misread their own digits, such as 0 and 6 or 1 and 7, and lose marks. Must they still write everything out?

A. Yes. Writing is still better. But I am not asking for beautiful handwriting. The point is to make it legible and keep the lines aligned. Even using lined paper and placing the equals signs (=) directly beneath one another will sharply reduce mistakes caused by misreading. Poor handwriting itself is not the problem. Work scattered in every direction is.

In Closing: When the Hand Moves First, the Mind Follows

The difference between students who are good at math and those who struggle is already visible in the first minute after they open an exam paper. One stares blankly at the question. The other uses a pencil to mark the conditions and cuts a road through it. It looks like a difference in talent. In truth, it is a difference in habit. And habits can be changed.

Once you have learned to solve by hand, the next problem is time. However well you solve, it means nothing if you cannot finish within the exam. The next chapter is about mastering the fifty minutes of an exam: how to command the clock instead of being chased by it.

Keep Your Error Notebook Fast, Not Pretty

A great many students exhaust themselves making an error notebook and give up. There is usually one reason: they are trying to make it too pretty. Copy the question out in several colours and transcribe the entire worked solution, and a single entry takes more than ten minutes. In truth, they never open that beautiful notebook again. They spent all their energy making it.

The error notebook I recommend has exactly two lines. On the first, write “why I got it wrong.” On the second, write “what I will do next time.” Cut out and paste in the question, or simply mark it on the exam paper; that is enough. For example: “Misread the condition and assumed $D > 0$ / Next time, circle the condition first.” Nothing more is needed.

The point is not the look of the notebook. It is to catch the thought you had at the moment you went wrong. A pretty notebook is good for showing other people. A two-line notebook changes you. The less time you spend arranging it, the more time you can put back into solving problems.

Fifty Minutes: How Not to Be Chased by the Clock

Opening Story: The Student Who Knows Everything but Runs Out of Time

Not long after Hyunwoo formed the habit of solving by hand, he ran straight into the next wall. When the paper arrived, he worked diligently from the first question onwards, one by one. Yet whenever the bell rang, the last three or four questions were still blank. When I marked the paper, some of those blanks turned out to be easy questions he could readily have solved.

This was not Hyunwoo's problem alone. A student I taught some years ago, Minseo, was exactly the same. In fact, their grasp of the concepts was fairly solid. Ask them a question and they explained it fluently; their homework was meticulous. Yet every exam ended with the final few questions untouched. "Maybe I just don't have a head for math. If I know the material, why do I keep running out of time?" they asked dejectedly. Rather than have them bring me the marked paper, I told Minseo to record the order in which they had solved the questions, together with the time spent on each. The next day's record made the cause obvious at a glance. They had spent a full eleven minutes on Question 4. Had they simply pushed those eleven minutes to the end, they could have answered three more easy questions correctly. What Minseo lacked was neither ability nor a head for math.

"I don't have enough time" is only half true. What Hyunwoo and Minseo lacked was not time but a **method for using it**. This chapter is not about methods of calculation. It is about **time management**: how to turn the same ability and the same fifty minutes into more marks.

1. The Real Reason You Run Out of Time: Management, Not Ability

The Trap of Solving in Order from the Front

Exam papers are generally arranged from easier questions at the front to harder ones at the back, but not always. One unusually awkward question is often planted in the middle. A student who solves only in order gets stuck on that one question and spends five minutes, then seven. Meanwhile, the questions farther back that they could have solved are washed away by the clock.

The Sunk-Cost Trap of Clinging to One Question

“I’ve come this far; it would be a waste to give up.” You have already spent three minutes, so it feels as though just a little longer will finish it. But that “little longer” consumes another three minutes. The time already spent will not return. You have to ask yourself, with a cool head: “Is this one minute best spent on this question, or on the two questions waiting behind it?”

I call what follows an “exam-room blackout.” You pour ten minutes into Question 14, suddenly look up at the clock, and discover that fifteen minutes remain. There are still five or six questions to solve.

At that instant your mind goes blank, cold sweat runs down your back, and even the easy questions that made sense moments ago begin to look like letters floating on the page. The sunk-cost trap is truly frightening because it does not merely steal time. It paralyses what remains with fear.

The Tragedy of Leaving Certain Marks Behind

The most painful marks are lost not because you answered incorrectly, but because you never saw the question. You could have solved it, yet ran out of time before you even touched it. That is a problem of management, not ability, which is precisely why it can be fixed.

I had a student like that too. In ordinary practice, this student worked at Grade 2 level, but every exam produced a Grade 4. When I inspected their papers, I found that they routinely spent nearly twenty minutes clinging to one difficult constructed-response question, then left four or five multiple-choice questions at the back completely blank. What they needed was not more study. They needed one change of order: take the questions you can solve first. As soon as they changed the order, they recovered Grade 2 in the very next exam. Their ability had not risen. The ability they already possessed had finally turned into marks.

2. When the Paper Arrives, Draw a Map First

Use the First Minute to Scan the Whole Paper

Do not charge at Question 1 as soon as the bell rings. Spend the first minute turning through the whole paper and reading the terrain. Where is it easy, where is it steep, and where are the high-mark questions? Two people can walk the same road at very different speeds depending on whether they know the map.

Take the Easy Questions and the Certain Marks First

Once you have scanned the paper, choose your order. Take the questions you can certainly solve and bank those marks quickly. This does two things. First, it prevents the tragedy of missing an easy question for lack of time. Second, banking marks early settles the mind, so you are less shaken when you reach a difficult question.

Budget Time by the Marks Available

If an exam is worth 100 marks and lasts fifty minutes, each mark is worth roughly thirty seconds. Spending five minutes on one four-mark question is the same as surrendering several three-mark questions you could have secured in that time. Marks are your time budget. There is no worse trade than clutching one difficult four-mark question and losing three easy questions worth three marks each.

Let us make the map more concrete. For a fifty-minute exam, I have students design a scenario in advance: “Set aside the two multiple-choice questions that look hardest and the final constructed-response question. Use the first thirty minutes for the initial scan and a first pass through everything else. Spend the next fifteen on the three questions set aside, and keep the final five for checking and marking.”

When the budget is allocated in advance, a difficult question cannot shake you: you know that a fifteen-minute reserve is still waiting. That is the difference between being chased by time and deploying it.

3. Three Rules for Commanding the Clock

Rule 1: The Two-Minute Rule — Mark It and Move On

If your hand has not found a way into a question after about two minutes, mark it with a star and move on. This is not retreat. It is regrouping. As you solve other questions, the clue to the one you left will often appear without warning. Returning after a break can help you look beyond the approach you first became fixed on.

Rule 2: The Two-Pass Strategy

On the first pass, solve only the questions you are sure of, quickly. Skip anything starred. Return to the stars on the second pass. You will then have secured every question you can solve, leaving all the remaining time to invest in the difficult ones. You are no longer being chased. You are in command.

Rule 3: Reserve the Final Five Minutes for Checking and Marking

If you cling to a new question until the bell, you may fail to transfer your answers or let mistakes slip through without a check. Keep the final five minutes empty, without exception, and use them to check the answer sheet and your calculations. This is when you use the “tracing back” habit from Chapter 2. Even after solving the whole paper, shifting every answer by one row while transferring them is more common than you might think.

One more point: these three rules cannot be switched on the morning of the exam. You must practise the two-minute rule and the two-pass strategy with a timer whenever you work through a problem book, so that your hand follows them automatically in the exam room. Time management is training too, and training belongs to the person who did it in advance.

Once Hyunwoo began using these three rules, the blank pages disappeared. His ability had not grown. He had begun to transfer all the ability he already possessed onto the exam paper.

ACTION CHECK

Three Things to Change Today from This Chapter

1. **Use the first minute after receiving the paper to scan the whole thing.** Do not charge straight at Question 1.
2. **If you are stuck on one question for two minutes or more, star it and move on.** Return on the second pass.
3. **Always reserve the final five minutes for checking and marking.**

WORKSHEET

Your Time-Management Scenario for the Next Exam

Do not improvise with time in the exam room. Fill this in beforehand and prepare your “operation.”
You can adapt it to the real exam simply by changing the numbers.

SAMPLE ENTRIES

Example (a fifty-minute, 100-mark exam):

- Initial scan and first pass through the certain questions → finish within **thirty minutes**.
- Questions to set aside: **(two hardest multiple-choice questions + final constructed-response question)**
- Time invested in the questions set aside: **fifteen minutes**
- Final check and marking: **five minutes** (keep these five minutes whatever happens)

My operation for the next exam:

Subject / total time / total marks _____

First pass, certain questions first (within how many minutes?) _____

Questions to set aside for now (type and number) _____

Time to invest in the questions set aside _____

Time reserved for the final check and marking _____

One Page for Parents

Q. My child says, "I ran out of time and couldn't finish." Should I make them practise solving faster?

A. Check the order before you worry about speed. Many students who run out of time are not slow with their hands; they lose the easy questions while clinging to one hard one. Ask, "Did you miss the easy ones because you were holding on to something difficult?"

Q. My child handles problem books well in ordinary practice, but the exam marks are always low.

A. The problem may be time management rather than ability. Even at home, have your child do "exam-condition practice," solving against the actual exam time. Training to solve under pressure from the clock uses a completely different muscle from training to solve a problem.

Q. How should we time practice at home?

A. Set the timer for five minutes less than the real exam: forty-five minutes for a fifty-minute paper. Use the remaining five minutes to practise checking and marking the answer sheet (OMR). Adjust the plan to the school's actual exam duration and answer format.

Q. My child always comes home from an exam with the last part unfinished.

A. The habit of solving strictly in order is the likely cause. They become trapped by a difficult question near the front and lose all the easy questions at the back. Have your child try it just once: "This time, put a star by the hard ones, move on, and come back later." You will see the blank pages at the end shrink sharply.

Q. My child is upset because there are so many questions they "got wrong through a silly mistake."

A. Many mistakes come not from a lack of ability but from having no time to check while being chased by the clock. Without the habit of reserving the final five minutes for checking, even questions your child knows will slip away. If mistakes are frequent, examine time management before increasing the volume of study.

In Closing: Time Is the Vessel That Carries Ability

However strong your ability, it cannot become marks unless you can fit it inside the fifty minutes called an exam. Time management is a rule, not a talent, and a rule can be changed today. As Hyunwoo discovered, blank pages are far more often the result of not knowing the right order than of lacking ability.

You now know how to solve by hand (Chapter 2) and command the clock (Chapter 3). One thing remains. However well you solve and manage the paper, not knowing what will appear still creates anxiety. The next chapter shows how to turn that anxiety into confidence: how to dissect past papers and read the examiner's patterns.

Past Papers: What to Do After Solving

Opening Story: Two Students Who Worked Through the Same Past Papers

Some years ago, two students in the same class, Jiho and Taeyun, each worked through the same collection of their school's past papers three times. They completed roughly the same number of questions and spent roughly the same amount of time. But Jiho focused each time on getting the answer right, while Taeyun asked of every question, "Why was this set?" From the outside, Jiho seemed to work harder: his rounds were faster and his tally of correct answers higher. Taeyun would hold on to one question and puzzle over it for so long that progress looked painfully slow.

Three months later, in the exam, the two reacted completely differently to an unfamiliar question. Jiho panicked: "This wasn't in the past papers." Taeyun calmly put pencil to paper: "Ah, they've twisted that concept this way." They had seen the same papers, but Jiho had memorised the questions while Taeyun had read the examiner. From that day, I had Jiho follow Taeyun's method. At first came the complaint, "If I go this slowly, when will I ever finish the material?" But after solving an unfamiliar question unaided in the second exam, Jiho never returned to the old way.

This chapter is not about how many times to solve past papers. It is about how to take them apart.

1. The Work Continues After You Mark a Past Paper

Many students use past papers like another problem book. They solve, mark, move on when they are right, and read the worked solution when they are wrong. Used this way, past papers are nothing more than a bundle of questions. Once solved, their useful life is over.

But the real value of a past paper lies not in the question itself, but in the **examiner** behind it. With school exams in particular, the teachers setting the paper are known, and the concepts they value, the types they return to, and the traps they favour are all preserved in past papers. A past paper is the most honest trailer for what will appear on the next exam, and how. Close it after checking the answer and you have left the cinema without watching the trailer.

2. Got It Right? Dissect It All the More

Most students revisit only the questions they got wrong. Top students dig even more fiercely into **the questions they got right**. They ask three things.

① Was It Ability, or Was It a Guess?

A question answered by guessing or by a vague feeling is not “one you got right.” It is one you simply have not got wrong yet. Twist it even slightly next time and you will. Even for a correct answer, check that you can explain, “This is the answer, for these logical reasons.”

② Was There a Faster Route?

If the answer was right but took five minutes, that solution is a “dangerous correct answer” in the exam room. The habit of looking for another route that solves the same question in two minutes attacks at its root the time problem discussed in Chapter 3.

③ Why Did the Examiner Give This Condition? Read the Pattern

This is the heart of it. We extend the “tracing back” of Chapter 2 across an entire set of past papers. Gather three years of one school’s papers and you will see the same tool returning each year in a different face. Suppose one year asks for the sum of the two roots of a quadratic equation, the next for their product, and the next for the distance between them. The three may look different, but all grow from one tool: the relationship between roots and coefficients. A student who reads that pattern walks into this year’s exam already prepared for whatever new face that tool may wear.

We need not look far for an example. The questions we worked through together in Chapter 2 are precisely like this. One asked for the distance between the two roots of $x^2 - 4x + 3 = 0$ (the roots 1 and 3, a distance of 2). Another asked for the value of k that makes the distance between the two roots of $x^2 + 2x + k = 0$ equal to 4 ($k = -3$). We solved the first by factorising and the second through the relationship between roots and coefficients. The first can also be approached through the sum and product of the roots, using $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$.

If one year’s paper asks for the sum of two roots, another for their product, and another for their distance, those are not three different questions. They are one tool appearing three times in disguise. The Chapter 2 examples let you practise making that connection. When you study actual past papers, look beyond appearances to the concepts used together.

3. Classify Your Wrong Answers: There Is More Than One Way to Be Wrong

Simply solving a missed question again is only half a review. What matters is **why you got it wrong**, and not all wrong answers are the same. Divide them into at least four categories.

- **Concept gap** — You did not know it at all. → Return to the concept and learn it again.
- **Calculation error** — You knew it, but slipped in the calculation. → Correct it with the habits of writing out the working and checking it from Chapter 2.
- **Missed condition** — You failed to see a condition such as “distinct” or “positive.” → Correct it by circling conditions, as in Chapter 1.
- **Ran out of time** — You could have solved it, but had no time. → Correct it with the time management from Chapter 3.

Each may be a “wrong answer,” but each needs an entirely different prescription. Try to cure a calculation error with more concept study and you merely waste time. Dismiss a concept gap as “just a mistake” and you will get it wrong again. **Classify the reason accurately and half the review is already done.**

I once classified a student’s exam paper in precisely these four ways.

- Could not find the maximum of a quadratic function → did not know the concept of the vertex itself → **concept gap**
- Set up the discriminant correctly, then copied a sign the wrong way round → **calculation error**
- Missed the condition “two distinct roots” and solved using $D \geq 0$ → **missed condition**
- Could readily have solved the question but ran out of time before touching it → **ran out of time**

All four looked identical under the heading “wrong,” yet their prescriptions were completely different: review the concept, build the habit of checking, circle the conditions (Chapter 1), and manage time (Chapter 3). Had we lumped them together as “wrong, so solve it again,” that student would have fallen in exactly the same places in the next exam.

The Right Form for an Error Notebook

An error notebook is not a place to copy out missed questions beautifully. A notebook that takes thirty minutes just to transcribe a question is close to a waste of time. You need exactly two lines: **“Why was I wrong? (Which category?)”** and **“What will I change next time?”** An error notebook with those two lines is more powerful than one in which you have copied the question a hundred times.

Place two students' error notebooks side by side and the difference is stark.

[A Typical Error Notebook] Cut out the missed question and paste it in, then use three coloured pens to copy fifteen lines of the worked solution beautifully. — Time: twenty minutes. What remains: the satisfying feeling of having studied.

[An Effective Error Notebook] "Cause: copied a sign the wrong way round (calculation error) / Prescription: compare the signs in the copied expression with the original." — Time: two minutes. What remains: a check to use on the next solution.

A pretty error notebook gives you the feeling of having studied. What changes the mark is usually the ugly notebook with two lines.

ACTION CHECK

Three Things to Change Today from This Chapter

1. **Do not "solve and mark" past papers; "solve and analyse" them.** Explain why even a correct answer is correct.
2. **Classify each wrong answer into four types: concept, calculation, condition, or time,** and write down the prescription that fits.
3. **Do not copy the question into your error notebook. Leave only two lines: "Why was I wrong? + What will I change?"**

The Two-Line Error Notebook: A Classification Sheet

Do not make a beautiful copy of the question you missed. Two lines, “cause (category)” and “prescription,” are enough. Study the example first, then fill in three recent wrong answers.

SAMPLE ENTRIES

- **Cause:** set up the discriminant correctly, then copied a sign the wrong way round → category: **calculation error**
- **Prescription:** next time, highlight the leading sign first and write each line in alignment.

Four categories (Chapter 4): ① concept gap · ② calculation error · ③ missed condition · ④ ran out of time

My wrong answer 1

Cause (+ category ①-④) _____

Prescription (what will I change next time?) _____

My wrong answer 2

Cause (+ category ①-④) _____

Prescription (what will I change next time?) _____

My wrong answer 3

Cause (+ category ①-④) _____

Prescription (what will I change next time?) _____

One Page for Parents

Q. My child puts enormous care into an error notebook, but the grades do not rise to match.

A. Check whether the time is being spent making the notebook pretty. Copying the question exactly and decorating it with coloured pens may look like study, but the actual effect is weak. Ask one thing: "Why did you get this question wrong?" If your child can give the reason unaided, the review is already complete, however plain the notebook looks.

Q. How many times is enough for working through past papers?

A. The method matters more than the number. Solve them three times while merely checking answers and nothing remains. Solve them once while asking "why was it set this way?" and the examiner's pattern remains. When your child starts saying, "So this school likes questions like these," they are using past papers properly.

Q. My child tries to move on by saying, "I only got this wrong by mistake."

A. For a student, the word "mistake" is the sweetest and most dangerous form of self-comfort. First distinguish a genuine slip in calculation from confusion caused by a weak concept. Say, "If it was a mistake, write exactly one line on why it happened." If your child cannot write that line, it was not a mere slip. It is a weakness still in hiding.

Q. Past papers from the school are difficult to obtain. What should we do?

A. In truth, the best past papers are already in your child's hands: the last midterm and final papers they sat. Before hunting down somebody else's papers, have your child dissect their own previous exams. The habits of that school, and of that teacher as an examiner, are already there in full.

In Closing: Past Papers Are a Window onto the Future

Past papers are not dead questions. They are a window illuminating the exam to come. Solve them to get the answers and they become a problem book you use once and throw away. Take them apart to read the examiner and they become a map of what will appear, and how.

We now have the skills for approaching a problem (Chapter 2), commanding the clock (Chapter 3), and reading past papers (Chapter 4): the craft of sitting an exam. Yet every one of those skills stands on a single foundation: concepts. The next chapter goes into the real fork separating top students from everyone else — the difference between “knowing” a concept and being able to use it.

A SHORT PAUSE

The Focus Created by Five Minutes of Clearing Your Desk

When studying will not start, the problem is often the desk, not your head. Sit down among yesterday’s handouts, a problem book for another course, a phone, and a snack packet, and your mind is scattered before you begin. To focus on one thing, first remove from sight the things that compete for your attention.

I am not proposing a grand clean-up. For exactly five minutes before you study, keep only the book and notebook for the subject you are about to work on, and one pen, on the desk; move everything else out of sight. Put your phone not merely on silent but in another room. If you keep checking it, leave it somewhere you cannot easily pick it up during the study session.

Clearing the desk is really practice in clearing the mind. It is a ritual for telling yourself: “From now on, I will focus on this one thing.” When you are frustrated that your ability is not growing, try clearing the desk once before changing how you study. Surprisingly often, that is where the problem lies.

Knowing a Concept Is Not Using It: How Top Students Learn Concepts

Opening Story: “But I Know All the Concepts”

One day Hyunwoo came to me looking thoroughly aggrieved. “Sir, I really do know all the concepts. But the moment I enter the exam room, I can’t apply any of them.” He could answer every “concept check” question at the front of a problem book, but his hand stopped as soon as he reached the application problems at the back.

I asked him, “Explain what the maximum and minimum of a quadratic function are.” Hyunwoo recited the definitions fluently. So I asked again, “Then do you know why you need that concept in this question?” Hyunwoo could not answer.

Hyunwoo “knew” the concept, but he did not know how to “use” it. In Chapter 1 we looked at rebuilding a concept in your own words and deriving it by hand. This chapter takes the next step. It is about the power to go beyond knowing a definition and call that concept up for use on an actual problem.

1. A Concept Is Not a Point but a Net

The first mark of a student who cannot use concepts is that they memorise each one separately: quadratic functions over here, quadratic equations over there, quadratic inequalities somewhere else. But in the mind of a top student, all three form **one picture**.

Consider the graph of the quadratic function $y = x^2 - 4x + 3$.

- Ask where this graph meets the x -axis ($y = 0$), and you are asking for the two roots of the quadratic **equation** $x^2 - 4x + 3 = 0$: $x = 1$, $x = 3$.
- Ask for the intervals where the graph lies above the x -axis ($y > 0$), and you are asking for the solution of the quadratic **inequality** $x^2 - 4x + 3 > 0$: $x < 1$ or $x > 3$.

Three different names — function, equation, inequality — are in truth only three views of **one graph**. Weave concepts into a net like this and one problem opens several doors at once. A student who has memorised each concept separately, by contrast, cannot call up knowledge about equations when the

problem arrives wearing the face of a function. There is no road between the concepts.

2. Why You Cannot Call Up a Concept You Know

What is missing for a student who knows a concept but cannot use it in a problem is a **bridge of translation** between definition and question. Knowing the concept alone is not enough. You must also know **when it appears**.

Top students memorise a signal alongside every concept. They pair a phrase in a question with the concept it should call up.

- “The quadratic graph is **tangent to the x -axis**” $\Rightarrow$ call up the discriminant condition $D = 0$.
- “**Maximum or minimum**” $\Rightarrow$ for a quadratic, check the vertex and domain. On a closed interval, compare the endpoint values too.
- “**Two distinct real roots**” $\Rightarrow$ call up the discriminant condition $D > 0$. (This is the condition we met in Chapter 1.)

This habit does not stop with grade-10 Common Mathematics. It becomes even more powerful in the courses that follow.

- In Calculus I, “the **gradient of the tangent** at a point on the curve” $\Rightarrow$ **differentiate** at that point (the derivative at the point).
- In Probability and Statistics, “**at least one**” $\Rightarrow$ translate it into the **complementary event**, subtracting the opposite case from the whole. (The “translation trap” from Chapter 1 was, in fact, a signal calling up a concept.)

This is the advanced form of circling conditions from Chapter 1. Do not stop at finding the condition; connect it to the concept it calls. When studying a concept, also write down an answer to “In what words will this concept appear in a question?” This trains you to recall the concept as you read the question.

3. How Top Students Check Their Concepts

Here are three ways to test whether a concept has reached the state in which you can use it.

① Retrieval on a Blank Page: Call It Up Without Looking

Close the textbook and write what you know about the unit on a blank page. A definition that looked familiar may not come back to you. Use this retrieval practice to find gaps, check them, and then try writing the ideas out once more.

And here the path of top students and everyone else divides once more. Not all blank pages are equal. One may be scattered with formulae; another may have them woven together with arrows. On the second, functions, equations, and inequalities are connected, with words such as “therefore” and “in other words” laid between them. Your true ability lies not in what you have written, but in how tightly it is woven.

② Build a Counterexample: Test the Boundary of the Concept

For a domain of all real numbers, is the statement “a quadratic function always has a minimum” true? It is true when the graph opens upwards (when the coefficient of x^2 is positive). But if it opens downwards, it has a maximum and no minimum. Ask yourself, “Is there a case in which this statement fails?” and search for a counterexample. Then you no longer know the concept vaguely; you know its exact boundary. The examiner’s traps always hide along that boundary.

③ Draw a Concept Map: See the Net

After finishing a unit, spread all its concepts across one page and connect them with arrows: function → equation → inequality. This map is the “net” from Section 1. If you can draw the map, the unit is yours. Any part you cannot draw is the place to study next.

Let us draw it a little more concretely. A concept map for the grade-10 unit on quadratic functions might look like this. In the centre, write “ $y = ax^2 + bx + c$,” then send out three arrows. ▶ “set $y = 0$ → quadratic equation (two roots)”; ▶ “set $y > 0$ or $y < 0$ → quadratic inequality (range of solutions)”; ▶ “view it as a graph → vertex and axis, intersections with the x -axis.”

Then connect “ $D = b^2 - 4ac$ ” to the end of all three branches. The discriminant determines how many roots there are and how many times the graph meets the x -axis. (To complete the solution of a quadratic inequality, you must also read which way the graph opens, the direction of the inequality sign, and whether equality is included.) Spread it all across one page and you can see that equations, inequalities, and graphs, which seemed to be learned separately, are really branches from the single root called a quadratic function. The moment you can draw this picture yourself is the moment you have truly mastered the unit.

$$y = ax^2 + bx + c$$

$y = 0$
Quadratic equation
(two roots)

$y > 0 / y < 0$
Quadratic inequality
(range of solutions)

Graph
Vertex · axis
Intersections with the x -axis

$$D = b^2 - 4ac$$

Determines the number of roots and intersections

ACTION CHECK

Three Things to Change Today from This Chapter

1. **Connect concepts instead of memorising them separately.** Bind functions, equations, and inequalities together on one graph.
2. **Write a “signal” beside every concept.** (For a quadratic, “touches the x -axis→ $D = 0$,” “maximum→check vertex and domain”)
3. **At the end of a unit, draw a concept map on a blank page.** The part you cannot draw is the weakness.

My Own Signal Table and Concept Map

Practise recalling the right concept by pairing the expressions in a question with the tools they call for. Study the examples, then fill in the signals for the unit you are studying now.

SAMPLE ENTRIES

- “The quadratic graph **is tangent to** the x -axis” $\Rightarrow$ discriminant $D = 0$
- “**Maximum / minimum**” $\Rightarrow$ check the quadratic’s vertex and domain; compare endpoints on a closed interval
- “**Two distinct real roots**” $\Rightarrow$ discriminant $D > 0$
- “The **gradient of the tangent** at a point on the curve” $\Rightarrow$ differentiate at that point (f')
- “**At least one**” $\Rightarrow$ translate into the complementary event

Signals for my unit: (expression from the question on the left; tool to call up on the right)

“ _____ ” $\rightarrow$ tool: _____

“ _____ ” $\rightarrow$ tool: _____

“ _____ ” $\rightarrow$ tool: _____

“ _____ ” $\rightarrow$ tool: _____

Draw a concept map: Put the central concept of this unit in the middle, then branch out with arrows to equations, inequalities, graphs, discriminants, and more. (Use the blank space below.)

One Page for Parents

Q. My child says, "I know all the concepts, but I can't apply them." Should I give them more application problems?

A. Before adding more questions, ask your child to explain the concept. Ask, "Why do you use that formula in this problem?" If there is no reason to give, the concept was memorised, not known. Usually, application fails not because there has been too little practice with application problems, but because there is no connection between concept and question.

Q. My child listens to lectures on concepts several times, but nothing changes.

A. Listening is "input." Real study happens during "retrieval." Rather than listening to the lecture one more time, it is far more effective to have your child write what they heard on a blank page unaided. If they cannot explain it, they do not know it.

Q. If my child follows every concept into the next and digs this deeply, won't studying ahead of the school curriculum slow down far too much?

A. Race ahead with a loose net and all those concepts will slip through by the final year. Weaving a tight net takes longer, but once it is properly made, your child will catch advanced problems incomparably faster. Look not at the speed of progress, but at the density of what remains.

Q. My child says the formulas are all memorised, but cannot use them in the exam.

A. Having "memorised" a formula and knowing "when to use it" are entirely different. For every formula, have your child write down, alongside it, "what words in a question tell me to use this formula?" — a signal of the kind described in this chapter. Only when a memorised formula connects to a problem does it become a weapon the student can call up and use.

In Closing: Connect the Concepts You Have Learned

Many students “know” concepts; few can “use” them. The difference is not a head for math. It is whether the concepts are left as separate points or woven into one net. Rebuild a definition in your own words (Chapter 1), connect concepts to one another, and learn the signals that call each one up in a question. Only then does a concept become a living tool.

We now know how to approach a problem and how to use concepts. In the next chapter we place these principles onto the actual courses. From Common Mathematics 1 and 2 through Algebra, Calculus I, and Probability and Statistics, I will point out the particular weak point to target in each.

A SHORT PAUSE

What Not to Do on the Night Before an Exam

The night before an exam is not the time to raise your ability, but to protect what you have built. Yet many students use this final night to erode it instead. Anxiety sends them into a new problem book or keeps them awake wrestling with difficult killer questions. Get stuck on a hard question you have never seen before on the eve of the exam and what you gain is not knowledge but anxiety: “I still have so far to go.”

The work for the night before is already decided. Do not learn something new; lightly review what you already know. Leaf through the two-line error notebook you have kept and say aloud, once each, the definitions of the concepts you often miss. That is enough. This is time to confirm what your hand remembers, not to cut a new road.

Above all, sleep. The concentration you preserve in the exam room with a clear mind will protect far more marks than the few questions you gain by staying awake all night. Sometimes the best study on the night before an exam is the courage to turn out the light early.

Every Course Has a Different Weak Point: From Common Math to Probability and Statistics

Opening Story: “The Course Changed and I Fell Apart Again”

Hyunwoo returned to me in grade 11. He had climbed to Grade 2 in grade-10 Common Mathematics, but Algebra and Calculus I were shaking him all over again. “Sir, I’m following the method exactly. Why is it difficult again?”

The helpless look from a year earlier had returned to his face, but it had a slightly different quality. In grade 10 he had stopped because he did not know how to begin. This time, unfamiliar symbols — limits, logarithms — tripped him up one step earlier: “What on earth does this mean?” The habit of solving by hand was in place, but the definitions he needed in order to set his hand to work were hazy, so the first line would not come. If anything, I was glad. This was not a retreat. It was a sign that he had reached a higher wall.

That was only natural. Every course has a different character. The tools we have learned so far — circling conditions, solving by hand, weaving concepts into a net — are fundamentals that work everywhere. But every course has its own weak point. In this chapter, we identify the weak points in the grade-10 and grade-11 courses tested in school exams, and decide where to aim the tools from the earlier chapters.

1. Common Mathematics 1 and 2: The Root of Everything (Grade 10)

Common Mathematics 1 and 2, studied in grade 10, cover polynomials, equations and inequalities, counting, coordinate geometry, functions and graphs, and more. They may not look glamorous, but **every course that follows is built on them.**

- **The weak points are “interpreting conditions” and “connecting graphs to expressions.”**

Equations and inequalities are where the “one-line condition” trap from Chapter 1 appears most often. The function units are where the “function = equation = inequality” net from Chapter 5 shows its power.

- Concepts missed in Common Mathematics are needed again when you meet new material in grade 11. That is why the prologue asks you to check gaps in the foundations early. Common Mathematics is not a course to finish quickly. It is the course to secure most firmly.

In particular, matrices, restored in the 2022 revision, and the coordinate geometry in Common Mathematics 2 make it easy to miss calculation steps or spatial relationships if you only skim them by eye. For matrix multiplication, you must point to each entry and calculate it by hand. In coordinate geometry, you must draw the coordinate plane yourself and place the points and lines on it. The “visualise by hand” habit from Chapter 2 comes into its own in these units.

[Worked Example · Common Mathematics 1] Find all real values of k for which the quadratic equation $x^2 - (k + 2)x + (k + 5) = 0$ has a repeated root.

Follow the Five Stages of Decoding from Chapter 2 exactly.

1. **Mark the conditions** — “has a repeated root”; the target is k .
2. **Call up the tool** — repeated root $\Rightarrow$ discriminant $D = 0$. (This is a “signal” from Chapter 5.)
3. **Set up the equation** — $D = (k + 2)^2 - 4(k + 5) = 0$.
4. **Calculate** — $k^2 + 4k + 4 - 4k - 20 = k^2 - 16 = 0$, hence $k = \pm 4$.
5. **Check and trace back** — When $k = 4$, $x^2 - 6x + 9 = (x - 3)^2$; when $k = -4$, $x^2 + 2x + 1 = (x + 1)^2$. Both have repeated roots.

→ Answer: $k = 4$ or $k = -4$. The difficult part was not the calculation, but the single translation “repeated root = discriminant zero.” The moment you write that line by hand, everything else is basic arithmetic.

2. Algebra and Calculus I: The Threshold of Abstraction (Grade 11)

Grade-11 Algebra covers exponentials and logarithms, trigonometric functions, and sequences. Calculus I covers limits and continuity of functions, followed by differentiation and integration of polynomial functions. This is where many students first feel that “math is abstract.”

- **The weak point is precision in definitions.** This is the stage on which the “definition trap” from Chapter 1 unfolds in earnest. The conditions on the base and argument of a logarithm, what it means for a limit to “exist,” the three conditions for continuity: you must hold these definitions as sentences, not feelings.

- **Do not be frightened by new notation.** Unfamiliar symbols such as $\log$, $\lim$, and Σ may look difficult, but they are only conventions. Rebuild each definition in your own words and visualise it with a graph, as in Chapter 5, and the abstract becomes concrete. Draw the graphs of exponential and logarithmic functions yourself, without exception. For a sequence, list several terms by hand and see the rule with your own eyes.

If sequences feel strange and forbidding, substitute 1, 2, and 3 for n yourself and list the terms by hand. The moment the abstract symbol a_n spreads out before you as the concrete numbers 3, 5, 7, 9..., the hidden rule (“increases by 2”) becomes visible of its own accord. Looking at concrete values first makes an abstract expression easier to approach. This too is an extension of solving by hand from Chapter 2.

[Worked Example · Calculus I] For the function $f(x) = x^3 - 3x^2 + 2$, find the gradient of the tangent to the curve $y = f(x)$ at the point $(1, f(1))$.

1. **Mark the conditions** — “gradient of the tangent” at $(1, f(1))$.
2. **Call up the tool** — gradient of the tangent at a point $\Rightarrow$ derivative at that point, $f'(1)$. (This is the “gradient of the tangent $\rightarrow$ differentiate” signal from Chapter 5.) Because only the gradient is required, the constant term $+2$, which vanishes on differentiation, does not affect the answer.
3. **Set up the equation** — $f'(x) = 3x^2 - 6x$. (The constant term 2 differentiates to zero and disappears.)
4. **Calculate** — $f'(1) = 3 - 6 = -3$.
5. **Check and trace back** — The gradient is -3 . Since $f(1) = 0$, the point of tangency is $(1, 0)$.

$\rightarrow$ Answer: -3 . If you can translate “gradient of the tangent” into $f'(1)$, you know which calculation this problem requires. The eye that tells which of the given information — such as the constant term 2 — matters for the gradient and which does not: that is decoding one level higher.

3. Probability and Statistics: The Course of Translation (Grade 11)

Probability and Statistics consists of counting, probability, and statistics. Where other courses ask for accurate calculation, this one asks for **translation from a situation into an expression**.

- **The weak point is the “translation trap” from Chapter 1.** A condition leaks out at the very moment you move a situation written in words into an expression. For “at least one”, consider the complementary event. Check separately whether order matters and whether repetition is allowed.

Do not select a formula from “simultaneously” or “one after another” alone: establish which results count as different.

For example, choosing two students for a pair is different from choosing a president and a vice-president. In the second case, exchanging the roles gives a different result even with the same two students. What matters is how the problem distinguishes outcomes, not simply the order of the physical selection.

- **Diagrams and tables are the strongest weapons.** When counting cases becomes confusing, do not count in your head. Spread the possibilities out as a tree diagram or a table. The “visualise by hand” habit from Chapter 2 shines especially brightly in this course.

[Worked Example · Probability and Statistics] A fair coin is tossed independently four times. Find the probability of obtaining heads at least once.

1. **Mark the conditions** — four tosses; heads “at least once.”
2. **Call up the tool** — Do not attack “at least” head-on; translate it into the complementary event. (This is the “translation trap” from Chapter 1.) That is, (heads at least once) = $1 -$ (the probability of no heads at all).
3. **Set up the equation** — no heads = tails on all four tosses = $\left(\frac{1}{2}\right)^4$.
4. **Calculate** — $\left(\frac{1}{2}\right)^4 = \frac{1}{16}$, hence $1 - \frac{1}{16} = \frac{15}{16}$.
5. **Check and trace back** — There are $2^4 = 16$ outcomes in all, and only one in which every toss is tails.

→ Answer: $\frac{15}{16}$. Trying to count “at least” directly would have sent the cases branching in every direction. Turn it over once into the complementary event and the work ends in a single line. This shows how reading the condition accurately can simplify the calculation.

ACTION CHECK

Three Things to Change Today from This Chapter

1. Treat Common Mathematics not as “a course to finish quickly” but as “the root to secure most firmly.”
2. State the definitions of new concepts in Algebra and Calculus I as sentences, then draw their graphs.
3. In Probability and Statistics, translate before calculating. Spread the situation out in a diagram or table.

WORKSHEET

Self-Check: The Weak Point in Each Course

Check whether you can identify the weak point in the course you are studying now. Anything that stops you is the next place to reinforce.

Common Mathematics 1 and 2 (Grade 10)

- ☐ I have the habit of circling the “one-line condition” in a problem
- ☐ I connect functions, equations, and inequalities on a single graph

Algebra · Calculus I (Grade 11)

- ☐ I can state the conditions on the base and argument of a logarithm
- ☐ I know the three conditions for continuity as sentences, not feelings
- ☐ “Gradient of the tangent” calls up differentiation

Probability and Statistics (Grade 11)

- ☐ I can turn “at least” into a complementary event
- ☐ I distinguish “simultaneously / one after another” and whether repetition is allowed

Unchecked items → the weak points I need to reinforce:

One Page for Parents

Q. My child did well in grade-10 math, then began to struggle with grade-11 Algebra and Calculus I. Have they reached the limit of their talent?

A. The character of the course has changed, not your child's talent. Grade-11 courses demand abstraction for the first time. What is needed here is not more questions, but time to secure the definitions of the new concepts. The moment of frustration when your child feels that the grade-10 method no longer works is, in fact, the finest opportunity to take their mathematical depth up one level.

Q. Probability and Statistics looks easy. Why does my child keep getting it wrong?

A. Precisely because the calculations look easy, Probability and Statistics invites carelessness. Check not just the calculation, but also how your child translated the situation into an expression. Have your child put the problem back into their own words by answering, "What exactly is this question asking?" Check that explanation, then have your child carry it through to the calculation and a final check.

In Closing: The Courses Differ, but the Tools Are the Same

Every course has a different weak point, but the tools that strike those points are the ones we have already learned. Interpret conditions in Common Mathematics by circling them as in Chapter 1; secure the definitions in Algebra and Calculus I with the concept-study methods from Chapter 5; translate Probability and Statistics with the solve-by-hand method of Chapter 2. This is why a new course need not frighten you. The weapons are already in your hands. When the course changes, change only where you aim.

We now know the terrain of the courses tested in school exams. Yet many students and parents still ask, "If I work this hard on school exams, when do I prepare for the CSAT?" In the next chapter, I will dispel that old misconception. We will look at how preparation for these two different tests can connect.

From School Exams to the CSAT: A Connected Three-Year Plan

Opening Story: “If We Keep Working on School Exams, When Do We Ever Prepare for the CSAT?”

This was the worry Hyunwoo’s mother raised most often in our consultations. “Sir, if we keep on top of school exams, there’s no time even to touch CSAT study. Won’t he collapse on the CSAT in grade 12?”

Many parents treat school exams and the CSAT as two separate tests. Trying to prepare each one independently exhausts child and parent alike.

School exams and the CSAT differ in format and preparation. But the concepts you learn can serve both. Starting from the 2028 CSAT syllabus, we will look at what to prepare together and what needs separate practice.

1. The 2028 CSAT: Start from the Common Syllabus

The mathematics examination changes from the 2028 CSAT onwards. **The old system, in which candidates chose one elective from among advanced calculus, probability and statistics, and geometry, is abolished. Every candidate is examined on the same common syllabus: Algebra, Calculus I, and Probability and Statistics.**

The significance of this change is greater than it first appears.

- **The courses you study for grade-11 school exams — Algebra, Calculus I, and Probability and Statistics — are the CSAT syllabus itself.**
You can use this shared syllabus for both school study and CSAT preparation. The year in which each course is taken depends on the school curriculum.
- Calculus II and Geometry in the 2022 revised curriculum are not included in the material directly examined by the 2028 CSAT. The dilemma of choosing an elective has disappeared too.

There is no need to treat school study as unrelated to the CSAT. Learn the concepts thoroughly at school, then practise the CSAT's question format and time management separately with past papers and official sample questions.

2. How Grade-10 and Grade-11 School Exams Lead into the Grade-12 CSAT

Connecting school study with CSAT preparation gives the following three-year plan. Adjust when you study each course to your school's curriculum.

- **Grade 10, Common Mathematics 1 and 2: the root of everything.** The roots of equations, functions, and inequalities grow directly into grade-11 Algebra and Calculus I. If Common Mathematics is weak, the grade-11 courses shake from end to end. Every year, I see students who dismissed grade-10 Common Mathematics as “easy” collapse in grade-11 Calculus I. It is not because they cannot differentiate. The functions and equations underneath are weak. Alongside differentiation practice, check the functions and equations on which it depends.
- **Grade 11, Algebra, Calculus I, and Probability and Statistics: the CSAT syllabus itself.** Concepts learned deeply here become foundations for later CSAT problems. Practise using them in unfamiliar contexts too.
- The important point is that earlier material is used again. If a new unit stops you, revisit the earlier concepts its solutions depend on.

In grade 12, leave ample time to consolidate earlier work and apply it under exam conditions. Some schools teach particular courses for the first time in grade 12, so also check what you have not yet covered. **The CSAT is not an exam that begins in grade 12. It has already begun with the first midterm of grade 10.**

One warning must be made plainly here. Around grade 11, a certain kind of student always appears: “I’m giving up on school exams and going all in on regular admissions through the CSAT.” They call themselves “CSAT fighters.” Which admissions route deserves more weight depends on the student’s record and the target university’s plan. But focusing on the CSAT is no reason to neglect the concepts being taught at school.

Think about it. Algebra, Calculus I, and Probability and Statistics, learned for school exams, are the CSAT syllabus. Abandoning school exams means walking out, under your own power, from the basic training ground for the very material that will appear on the CSAT. Use the weaknesses revealed by school exams in your CSAT study too. Deciding which test deserves more time is a different question from learning the necessary concepts properly.

3. A Three-Year Roadmap That Does Not Wobble

Built on this structure, here is the focus for each year.

- **Grade 10: the year to plant roots and methods.** Treat Common Mathematics 1 and 2 not as courses to finish quickly but as the roots to secure most firmly. At the same time, make the methods from Chapters 1-5 — solving by hand, managing time, weaving a concept net — second nature. Habits formed here last for all three years.
- **Grade 11: the year to deepen the shared syllabus.** You study Algebra, Calculus I, and Probability and Statistics. Use school study to learn the concepts in the common CSAT syllabus, and give those exams everything while adding the past-paper analysis from Chapter 4 and viewing them through “CSAT eyes” too.
- **Grade 12: the year to consolidate and sharpen.** Check which parts of the syllabus you have not yet studied. Bring the concepts built in grades 10 and 11 up to exam speed and repair weak units. If the previous two years were weak, the year is consumed by repairs; if they were secure, you can leap into advanced work.

One tip for the High School Credit System: this is now an age in which students choose their courses. But there need be no confusion about mathematics. Whatever your intended field, humanities or sciences, Algebra, Calculus I, and Probability and Statistics are the common CSAT syllabus and **courses everyone must be sure to take**. If you aim for medicine or the very top science programmes, however, Calculus II and Geometry, although not directly examined on the CSAT, may matter as completed courses on the school record or as recommended preparation for the intended major. Whether a university considers them in its own written examinations or interviews must be checked separately in that university’s admissions plan. It is worth taking them selectively, according to where you aim to go.

ACTION CHECK

Three Things to Change Today from This Chapter

1. **Drop the idea of “school exams over here, the CSAT over there.”** The grade-11 school-exam courses are the CSAT syllabus.
2. **Treat grade-10 Common Mathematics as the starting line for the CSAT.** Repair gaps when you find them.
3. **From grade 11, view school exams through “CSAT eyes” too.** Add the past-paper analysis from Chapter 4 to your school-exam study.

WORKSHEET

Draw My Three-Year Roadmap

Connecting school study with CSAT preparation, write down where you stand now and what deserves your focus. The common mathematics syllabus for the 2028 CSAT is Algebra, Calculus I, and Probability and Statistics.

SAMPLE ENTRIES

- **Grade 10 — the year to plant roots:** treat Common Mathematics 1 and 2 as “the roots to secure most firmly.” Make the methods in Chapters 1-5 habitual.
- **Grade 11 — the year to deepen the shared syllabus:** view Algebra, Calculus I, and Probability and Statistics through “CSAT eyes” too.
- **Grade 12 — the year to consolidate and sharpen:** check unstudied material. Repair weak units and raise everything to exam speed.

Where I am now, and my plan:

I am in grade (); courses I am studying _____

What I will secure most firmly this semester _____

Method from this book I will make a habit this year _____

Unit that worries me most / reinforcement plan _____

One Page for Parents

Q. The CSAT is changing again. I have no idea how to prepare my child.

A. This change is actually welcome news for parents. With electives gone, there is no longer a dilemma over what to choose, and the courses learned for school exams are now the CSAT syllabus. Help your child learn the concepts deeply at school, and practise the CSAT's question format and time management separately.

Q. My child is in grade 10. Do we really need to think about the CSAT already?

A. You need not worry about it. It is enough to know that grade-10 mathematics is the root of the CSAT. Not skimming over Common Mathematics now is already the greatest investment in the CSAT three years ahead. There is no need for haste, but this is not a period to treat lightly.

Q. My child intends to follow the humanities track. Must they still take Calculus I?

A. Yes. They must. Beginning with the 2028 CSAT, Algebra, Calculus I, and Probability and Statistics form the common syllabus regardless of humanities or sciences. Dropping Calculus I because your child is in the humanities is like leaving one entire pillar of CSAT mathematics empty. Every student must secure all three courses, whatever their intended field.

Q. My child is already in grade 12. Is it too late to change the method now?

A. It is not too late. Check the time remaining and any unstudied material, then start repairing the weakest unit with the methods in this book. Using the remaining time with the right method is always better than giving up because it feels late.

In Closing: Three Years Are One Road

See school exams and the CSAT as two mountains and you will be chased for three years. See them as one road and the direction becomes clear. The common syllabus of the 2028 CSAT provides a basis for connecting preparation for both tests. Grade-10 roots, the grade-11 syllabus, grade-12 depth: these are not three disconnected tasks, but one journey along a single line.

We have now mapped everything from why students collapse (Chapter 1) through solving, time, past papers, concepts, strategies for each course, and the three-year road (Chapters 2-7). One last subject remains: the strength to carry all of this through. Mindset. In the next chapter, we will talk about how to stand up after ruining the first exam, and about the parent's role beside the student.

A SHORT PAUSE

A Slump May Be a Sign That Your Ability Is Growing

When grades stay still or even fall despite hard work, students often collapse into the thought, "Maybe this is as far as I go." But over many years I have often seen that plateau arrive just before ability leaps to the next level. During the transition from solving by eye to solving by hand, and from memorising problem types to understanding concepts, the familiar old method is breaking down while the new one has not yet settled. For a while, the grades wobble.

Picture a staircase. It is not a diagonal slope but a series of flat stretches alternating with vertical rises. Ability grows the same way. For a time, nothing seems to change; then suddenly you step up a level. Give up on the flat stretch and you never reach the next step.

So when a slump arrives, first check whether the method is wrong. But if you are confident that the method is right, hold on a little longer. Today's frustration may not be retreat, but the run-up to your next leap.

Getting Back Up on the Day You Fall: Mindset for Students and Parents

Opening Story: The Day Hyunwoo Received Grade 4

In the first chapter of this book, Hyunwoo received Grade 4 on his first high-school midterm. The story I will tell now is, in fact, the story of that evening: the evening he received the shocking report card. Before he could change how he studied, the first thing Hyunwoo had to overcome was not a thick problem book, but the loss of confidence to begin again.

That day, Hyunwoo closed his bedroom door and did not come out for a long time. “Maybe I simply don’t have a head for math.” It was the moment when he was ready to call everything a matter of talent and give up. Dinner went cold on the table. His mother later told me that she had paced outside the closed door several times, then simply walked away because she did not know what to say. “If I pushed him to study more, I thought he might break. If I said it was all right, that felt like a lie...” That was how she had felt that night.

In truth, the most frightening thing that night was not the number Grade 4. It was the label taking shape in a parent’s mind: “What if my child truly cannot do math?” Had Hyunwoo actually given up then, none of the methods in this book would have mattered. Every method of study works only on top of a willingness to try again.

This chapter is therefore the last in the book and yet, in truth, the story that is needed first. It is about how to rebuild a spirit that has collapsed.

1. A Report Card Is Not a Verdict but a Diagnosis

Do badly on the first exam and it feels as though the world has ended. Under the five-band system, especially, one mistake can appear to move a grade sharply. But a report card is not a verdict declaring, “This is as far as you go.” It is a diagnosis saying, “This is where you are now, and this is the way forward.”

A doctor reads a diagnosis and writes a prescription instead of sinking into despair. Read a report card in the same way. Instead of being crushed by a Grade 4, ask, “Where and why did I go wrong?” using the classification from Chapter 4. At that moment, the report card ceases to be grounds for despair and becomes a blueprint for the next exam. And remember: the first exam of grade 10 is only the first five minutes of a long, three-year match. No match ends because you conceded a goal in the opening five minutes.

2. Trust the Process, Not “Talent”

At the root of a collapsed spirit lies a misunderstanding called talent. The belief that “you need a good head to do well at math” turns one failure into an apparent verdict on talent. From beginning to end, this book has tried to prove the opposite. **Math is method, not talent, and method enters the body through a process.**

Trust the process, then, not the result. Think of strength training. Lift a barbell today and you will not see new muscle in the mirror tomorrow morning. Work through math with the right method today and, likewise, it will not show up in tomorrow’s marks. Even before the marks change, check whether you can now explain an idea or complete a solution that stopped you yesterday. One problem written out by hand today, one question’s intent traced back today: even if neither appears immediately as a mark, that time has not been wasted.

Change whom you compare yourself with, too. Not the student beside you, but the person you were yesterday. If you understand one concept more precisely than you did yesterday, that day’s study has succeeded. These small successes accumulated, unnoticed, until they turned Hyunwoo’s Grade 4 into Grade 2. It was not a miracle. It was the change created by time spent trusting the process.

3. The Parent’s Role: Pacesetter, Not Supervisor

From here, I am speaking to parents. Your role in your child’s mathematical journey is decisive. But it must be the role of a pacesetter, not a supervisor.

Surveillance and pressure alone do little to help a child learn to work independently. Instead, become the pacesetter alongside your child, matching their breathing. The specific method is exactly what I have repeated at the end of each chapter. Ask not “What mark did you get?” but “Will you explain just one of the problems you solved today?” Ask about the process, not the mark.

Suppose your child has just had an exam go badly. “After all the time you’ve spent at the academy, how can these be your marks?” is the worst thing you can say: it turns the result into a weapon of guilt. Ask this instead: “What was different between the problem book you worked through at the academy and

the questions on this exam?” Both questions concern the exam. The first closes your child’s mind; the second lets them analyse the cause for themselves.

Above all, do not load your own anxiety onto your child. A child reads their future in a parent’s expression. Only when the parent first believes, “It’s all right; we can change the method,” can the child pick up the pencil again. What brought Hyunwoo back to his desk that evening was not, in truth, a method of study. It was the sentence, “Let’s try again.”

ACTION CHECK

What to Hold On To from This Chapter (for Students)

1. **A report card is not a verdict but a diagnosis.** Do not collapse under the number; read the “why.”
2. **Do not compare yourself with others. Compare yourself with who you were yesterday.**
3. **Trust the process even when the result is invisible.** The one problem you solve today is not wasted.

WORKSHEET

Compare Yourself with Yesterday

The person to compare yourself with is not the student in the next seat, but the person you were yesterday. Comparison with others leaves you helpless; comparison with yesterday gives you the strength to keep moving. Write exactly three lines today.

SAMPLE ENTRIES

- **One thing that went better than yesterday:** Today I solved, unaided, a type of quadratic inequality I could not solve yesterday.
- **Problems I wrote out by hand to the end today:** 4
- **A word to myself tomorrow:** I may be slow, but the direction is right. Keep going.

Today’s record:

One thing that went better than yesterday _____

Problems I wrote out by hand to the end today _____

A word to myself tomorrow _____

Not the number on the report card, but the accumulation of these three lines: that is the real evidence that ability is growing.

One Page for Parents

Q. What should I say to a child devastated by a bad first exam?

A. “It’s all right. This is only the beginning. You do not lack a head for math; you simply haven’t changed the method yet.” Begin by acknowledging how your child feels. When they are ready to talk, listen to what was difficult and look together for something they can change.

Q. My child wants to give up math altogether.

A. The helpless belief “Nothing I do will work” may be part of it. First listen to what your child is finding difficult. Instead of presenting a grand goal, create one very small success: help your child solve today one problem they could not solve yesterday.

In Closing: In the End, the Spirit Comes First

In this book we have discussed a great many methods, from how to solve a problem to a three-year roadmap. But every one of them works only on a single foundation: the willingness to try again. Hyunwoo’s true beginning was not any method of study. It was the spirit with which he picked up his pencil again on the evening he had collapsed.

The methods are all here in this book. All that remains is to trust them and take the first step.

The Method Is Already in Your Hands

If you have read this book from beginning to end, most of the tools you need are now in your hands: circling conditions, solving by hand, dissecting past papers, connecting concepts, managing time, and spotting the traps in each course. Their names differ, but they all converge on the Five Stages of Decoding.

Remember this. Not one sentence in this book demanded “innate talent.” From beginning to end, it has spoken only of method and the willingness to try again. In a Cambridge lecture room, in an essay-based admissions examination against odds of more than a hundred to one, and at Hyunwoo’s desk after he received Grade 4 and closed his bedroom door, only those two things changed. Talent did not.

Hours of study may not turn straight into marks. But if you can solve a problem today that stopped you yesterday, that progress is yours. Whatever number is on your report card now, it is closer to a starting point than a limit.

Open a problem book tonight and solve the first question all the way through by hand, not by eye. Circle the conditions. Write what is given and what is required in the margin. That one question is your first step toward making the method in this book your own.

— Sungmin Kim

The 30-Day Math Reset: An Action Plan for Your Grade Band

If you do not know what to apply first from this book, or how, follow the thirty-day plan below exactly. Choose the one track that fits where you are now. Each track covers four weeks, roughly thirty days, with a goal and a set of actions for every week.

The grades here refer to the five-band system used for school grades under the 2028 framework. Choose your track less by the exact grade than by the place where you are stuck now.

Track A — Grades 1-2 (Top Band): “Eliminate Mistakes and Reach the Very Top”

Even strong students can have gaps in concepts. Identify the causes of lost marks first, then focus on solving familiar material reliably.

- **Week 1 — Map your mistakes:** Sort questions missed on recent school exams and mock examinations into the four categories from Chapter 4: concept, calculation, condition, or time. This week’s goal is to discover which categories account for your lost marks.
- **Week 2 — Retrace the questions you got right:** Following Chapter 4, find faster and more secure routes through questions you answered correctly. Train not for the answer, but for speed and stability.
- **Week 3 — Reverse-engineer three years of past papers:** Extract the tools that recur across three years of your school’s past papers, like the relationship between roots and coefficients in Chapters 2 and 4. Practise using the same tool under different conditions.
- **Week 4 — Optimise time + constructed responses:** Cement the time-management method from Chapter 3 through forty-five minutes of solving and five of checking and marking, and check the causes of lost marks in constructed-response questions one by one.

1. Track B — Around Grade 3 (Middle Band): “Turn Concepts into Weapons”

You seem to know the material, but cannot produce it in the exam. Bring each concept to a state in which you can use it.

- **Week 1 — Weave a concept net:** Following Chapter 5, draw a concept map for every unit you have studied. Connect functions, equations, and inequalities on one page.
- **Week 2 — Establish the Five Stages of Decoding:** Use the Five Stages of Decoding from Chapter 2 on five questions a day. Replace the habit of solving by eye with solving by hand.
- **Week 3 — Build signals + classify wrong answers:** Organise the “condition → concept” signals from Chapter 5 and sort missed questions into the four categories from Chapter 4.
- **Week 4 — Focus on one weak unit + practise for forty-five minutes:** Choose your weakest unit and secure it again from the concepts up. At the weekend, practise forty-five minutes of solving followed by five of checking and marking.

2. Track C — Grades 4-5 (Starting from the Foundations): “Back to the Roots, but Quickly”

What you need now is not ambition, but the right first step. Begin with exactly three habits.

- **Week 1 — Hand, not eyes:** Following Chapter 2, do only three questions a day, but write each one out completely by hand. Cover the worked solution and reach the point where you can solve it again unaided.
- **Week 2 — Retrieve root concepts on a blank page:** Write the central concepts of Common Mathematics — equations and functions — on a blank page in your own words. This is the retrieval training from Chapter 5.
- **Week 3 — Circle the conditions:** Build the Chapter 1 habit of marking the conditions in a question. Start slowly with easier past-paper questions.
- **Week 4 — Become conscious of time + one line on each error:** Develop the sense of time from Chapter 3, then organise each missed question with the two-line error notebook from Chapter 4.

3. Weekly Action Record (Complete It for Your Track)

Rewrite the four-week plan for the track you chose above — A, B, or C — in your own words in the table below, and mark each week when it is done. There is no need to make it grand. Record only the one thing you will do this week and whether you did it.

WORKSHEET

My Thirty-Day Record

Chosen track: (tick one of A · B · C) A ☐ B ☐ C ☐

Week	What I will do this week (in my words)	Done
Week 1		☐
Week 2		☐
Week 3		☐
Week 4		☐

One line on what changed after thirty days:

4. After Thirty Days

The aim of this thirty-day plan is not to promise a higher grade, but to review how you study and build a habit. Once the habit has settled, the mark has room to move. Do not rush. As Hyunwoo discovered, change begins at the point where the method changes.

5. After Four Weeks: Five Questions to Ask Yourself

After a month, being able to answer the five questions below matters more than the report card. Parents can check them with their child; students can check them alone.

ACTION CHECK

Thirty-Day Check

1. **Hand:** Does the margin of my latest exam paper contain “given” and “required” in my own handwriting?
2. **Signals:** For a question I got wrong, can I tell whether I missed the “one-line condition” or failed to call up the tool?
3. **Wrong answers:** Is my error notebook two lines of “cause + prescription,” rather than copied questions?
4. **Time:** Before an exam, do I plan forty-five minutes for solving and five for checking, or adjust that allocation to my own exam?
5. **Process:** Even if the mark is unchanged, is there at least one problem where I am one step less stuck than I was yesterday?

A Mini Dictionary of Easily Confused Terms and Symbols

This glossary collects the terms and symbols that appear often in this book and in school lessons, paying particular attention to the points of confusion that lead to mistakes in exams. Pinning a definition down in a sentence, rather than leaving it as a vague impression—that is where Grade 1 begins. (The terminology follows the 2022 revised national curriculum.)

Equations, inequalities, and functions

Direction of the inequality sign

When you multiply or divide both sides by the same number, reverse the inequality sign if that number is negative.

Caution Forgetting to reverse the sign when multiplying both sides by -2 is a common source of errors in Chapters 2 and 4.

Domain of a function

The set of values that may be substituted for x . When logarithms, fractions, and roots impose conditions at the same time, read the domain as the intersection of all those conditions.

Caution If you see only the word “defined” and fail to write down the conditions on the argument, denominator, and radicand one by one, you will go wrong immediately on a domain question.

Discriminant D

For the quadratic equation $ax^2 + bx + c = 0$ ($a \neq 0$), $D = b^2 - 4ac$. It determines the number of real roots. Here the quadratic coefficients are all real.

Caution If $D > 0$, there are two distinct real roots; if $D = 0$, there is a repeated root (two equal real roots); if $D < 0$, there are no real roots (there are two distinct non-real roots).

Repeated root

The case in which the two roots of a quadratic equation are equal. Then $D = 0$, and the graph is tangent to the x -axis.

Caution Do not confuse “two distinct real roots” ($D > 0$) with “has real roots” ($D \geq 0$).

Real and non-real roots

A root that is a real number is a real root; one that is not a real number is a non-real root.

Caution Remember that the condition “has real roots” is $D \geq 0$, including a repeated root.

Maximum and minimum values of a quadratic function

On a domain of all real numbers, the quadratic function $y = ax^2 + bx + c$ has either a maximum or a minimum value at its vertex.

Caution If $a > 0$, the parabola opens upwards and has only a minimum; if $a < 0$, it opens downwards and has only a maximum. “It always has a minimum” is false. On a closed interval, compare the values at any vertex within the interval and at both endpoints.

1. Algebra (indices, logarithms, and sequences)

Logarithm $\log_a b$

a is called the base, and b the argument.

Caution (conditions for it to be defined) The base must satisfy $a > 0$ and $a \neq 1$; the argument must satisfy $b > 0$. Miss these conditions and you will go wrong immediately on a domain question.

Common ratio of a geometric sequence

The constant multiplier that takes one term to the next. When the preceding term is non-zero, divide the next term by it to find the ratio.

Caution The formula for the sum of a geometric sequence splits into the case where the common ratio is 1 and the case where it is not 1. Students often forget this special case ($= 1$).

2. Calculus I (limits, continuity, and differentiation)

A limit of a function “exists”

As x gets arbitrarily close to a particular value, the value of the function approaches a fixed value.

Caution A limit exists only when both the left-hand and right-hand limits exist and are equal. If they differ, the limit does not exist.

Continuity of a function (at a point)

For a function $f(x)$ to be continuous at $x = a$, three conditions must hold at once—① $f(a)$ is defined; ② $\lim_{x \rightarrow a} f(x)$ exists; ③ and the two are equal ($\lim_{x \rightarrow a} f(x) = f(a)$).

Caution The vague impression that “it is continuous if it looks roughly connected” is a trap.

Derivative at a point $f'(a)$

The instantaneous rate of change of the function $f(x)$ at $x = a$. Geometrically, it equals the **gradient of the tangent** at the point $(a, f(a))$ on the curve $y = f(x)$.

Caution When you see the phrase “gradient of the tangent”, that is a signal to differentiate at that point.

3. Probability and Statistics

Complementary event

The event in which a particular event A does not occur. $P(A) = 1 - P(A^c)$.

Caution For phrases such as “at least one”, do not count head-on. Turn the question around and use the complementary event; most such problems can then be solved in a single line.

Permutations and combinations

If order matters, use a permutation; if it does not, use a combination.

Caution Read the wording precisely — whether it is “at the same time” or “in succession”, and whether repetition is allowed. Otherwise, you will take the wrong fork between permutations and combinations.

4. Geometry (coordinate plane and space)

Distance between two points

In the coordinate plane, if $P(x_1, y_1)$ and $Q(x_2, y_2)$, then $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Caution The difference between two roots found through the relationship between roots and coefficients, $|\alpha - \beta|$, can also be worked out using the same formula. The Chapter 2 example can be viewed as the distance between two points on a number line.

Equation of a circle

For centre (a, b) and radius r , the equation is $(x - a)^2 + (y - b)^2 = r^2$.

Caution In questions about “a point on the circle” or “the relative position of a circle and a line”, set up the equation first, then connect it to the discriminant or the distance formula.

Volume of a solid

Memorise the formulae for surface area and volume, and use a diagram to check “base area $\times$ height” for a prism or cylinder. A pyramid or cone has one third of the volume of a prism or cylinder with the same base area and height.

Caution Students often confuse units (square versus cubic), or swap the radius and the height in problems involving a solid of revolution.

The 2028 Decoding Map, Course by Course

This single-page map lays the tools you have learned in this book across the four courses you will often meet in grades 10 and 11. Open it before you begin each new course. At a glance, you will see where that course sets its traps, which signals call for which tools, and how every solution ultimately comes together in one routine: the **Five Stages of Decoding** (① Mark the conditions → ② Call up the tool → ③ Set up the equation → ④ Calculate → ⑤ Check and trace back).

※ **A note on when each course is taken:** Course arrangements vary by school, region, and academic track. The grades shown below refer to **arrangements commonly found in many schools**. Check your own school's curriculum first.

When to open this map

- In the first week of a new course or unit—decide in advance, “What is the trap in this course?”
- The day before an exam—rather than solving more problem types, lightly review only the links from **signal** → **tool**.
- When organising your error notebook—check for calculation errors as well as missed conditions or concepts you failed to call up.

1. One line of decoding for each course (before you open the table)

Common Mathematics 1 and 2

Trap: Missing “one line of conditions” (two distinct roots, all real numbers, a repeated root, and so on).

Signal: for a quadratic, “repeated root” or “distinct real roots” $\Rightarrow$ discriminant D ; a quadratic graph “touches the x-axis” $\Rightarrow D=0$.

By hand: Circle the conditions and draw the graph yourself.

First two weeks: Make a one-page signal chart for discriminants and vertices in the quadratic-functions unit.

Algebra

Trap: Omitting defining conditions (logarithm argument > 0 , common ratio $= 1$ in a geometric sequence, and so on).

Signal: “logarithm” $\Rightarrow$ base and argument conditions; “sequence” $\Rightarrow$ substitute $n = 1, 2, 3$.

By hand: Confirm the pattern with your eyes before moving to the general term or formula.

First two weeks: Write a “defining-conditions checklist” for logarithms and sequences on a blank page.

Calculus I

Trap: Confusing the definitions of limit, continuity, and differentiability.

Signal: “gradient of the tangent” $\Rightarrow f'(a)$; “continuous” $\Rightarrow \lim_{x \rightarrow a} f(x) = f(a)$.

By hand: Put the definition into a sentence, then match it to a graph.

First two weeks: Write the three conditions for continuity once each in words, symbols, and a diagram.

Probability and Statistics

Trap: Letting conditions leak away when translating words into symbols.

Signal: “at least” $\Rightarrow$ complementary event; “at the same time” or “in succession” $\Rightarrow$ distinguish the cases.

By hand: Do not count in your head; spread the possibilities out in a tree or table.

First two weeks: Translate five sentences containing “at least” or “at the same time” into complementary events or counts of cases.

2. The one-page summary table

Course	Typical trap	Signal (condition $\Rightarrow$ tool)	The key move on paper	One thing a parent can say
Common Mathematics 1 and 2	Missing “one line of conditions” (two distinct roots / all real numbers, and so on)	For a quadratic, “repeated root” or “distinct real roots” $\Rightarrow$ discriminant D / a quadratic graph “touches the x-axis” $\Rightarrow D=0$	Circle the conditions and draw the graph yourself	“Which condition did you circle in this question?”
Algebra	Omitting defining conditions (logarithm argument > 0 , geometric common ratio)	“Logarithm” $\Rightarrow$ base and argument conditions / “sequence” $\Rightarrow$ list the terms	Substitute 1, 2, 3 for n and see the pattern for yourself	“Can you explain why this symbol is defined this way?”

Calculus I	Confusing the definitions of limit, continuity, and differentiability	“Gradient of the tangent” $\Rightarrow$ differentiate (f') / “continuous” $\Rightarrow$ limit = function value	Define it in a sentence rather than by feel, then show it on a graph	“What exactly does ‘continuous’ mean?”
Probability and Statistics	Conditions leaking away when translating words $\rightarrow$ symbols	“At least” $\Rightarrow$ complementary event / distinguish “at the same time” from “in succession”	Do not count in your head; spread it out in a tree or table	“Can you tell me again exactly what this question is asking?”
Shared routine	Every course and every question	① Mark the conditions $\rightarrow$ ② Call up the tool ③ Set up the equation $\rightarrow$ ④ Calculate ⑤ Check and trace back	Circle the conditions · write by hand · check	“At which of the five stages did you get stuck?”

3. Checklist for the first two weeks of a new course

ACTION CHECK

With the decoding map

1. Memorise just **one trap**—choose “the one line students get wrong most often in this course” from the table and write it down.
2. Connect **two signals**—draw arrows from the condition in words to the tool it calls for (a formula or definition).
3. Solve **three past-paper questions** with the Five Stages of Decoding—one at a time, “by hand and all the way through”, putting completion before speed.

How to use it: When you meet a question, ① identify the course and its “typical trap” $\rightarrow$ ② use the “signal” to summon the right tool $\rightarrow$ ③ put your hands to work through the Five Stages of Decoding. Use this table to decide where to begin with an unfamiliar question.

4. A three-minute routine you can use on the spot

Just before an exam, on the first day of a new unit, or while organising your error notebook, spend three minutes repeating nothing but the sequence below.

ACTION CHECK

Three-minute decoding

1. **30 seconds**—say aloud one line describing the trap in this course.
2. **1 minute**—write two signals in the form “condition $\Rightarrow$ tool”.
3. **1 minute 30 seconds**—take one past-paper or textbook question and write by hand only stages ①–③ of the Five Stages of Decoding (save the calculation for later).